

MANITSOQ SUKKERTOPPEN, Greenland

WMO: 042410

Lat: 65.4110N

Lon: 52.9325W

Elev: 32

StdP: 100.94

Time Zone: -3.00 (GRL)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-20.1	-17.5	-25.9	0.4	-18.7	-23.7	0.4	-15.5	19.7	-2.7	16.2	-7.7	4.9	80	0.906

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	15.3	9.3	13.6	8.7	12.2	8.2	10.3	13.4	9.6	12.4	8.9	11.3	4.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.1	10.6	8.1	6.7	9.8	7.4	6.4	9.3	30.2	13.4	28.4	12.6	26.8	11.4	14.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.0	11.2	DB	-20.4	20.6	3.7	1.9	-23.1	21.9	-25.3	23.0	-27.3	24.1	-30.0	25.5	(4)
(5)			WB	-20.8	12.2	3.5	1.2	-23.3	13.1	-25.4	13.7	-27.4	14.4	-29.9	15.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-7.4	-7.6	-7.6	-2.9	1.7	5.7	8.7	8.2	5.1	0.9	-3.1	-6.0	(6)	
	DBStd	7.14	4.78	5.80	5.86	4.15	3.22	2.67	2.74	2.14	2.21	3.27	3.69	4.41	(7)	
	HDD10.0	3795	539	492	547	386	258	130	59	66	147	284	393	494	(8)	
	HDD18.3	6806	797	725	806	636	516	378	300	315	396	542	643	753	(9)	
	CDD10.0	31	0	0	0	0	1	2	18	8	1	1	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	4.9	5.6	5.2	4.9	4.7	4.4	4.4	4.2	4.3	5.2	4.7	5.4	5.3	(14)
(15) Precipitation	PrecAvg	590	57	36	35	36	32	36	47	65	66	56	64	61	(15)	
	PrecMax	822	157	86	92	81	58	75	91	109	142	117	134	128	(16)	
	PrecMin	392	5	14	3	13	8	12	18	23	19	16	24	26	(17)	
	PrecStd	106	35	17	21	18	14	15	21	22	26	22	27	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	7.2	7.6	9.6	12.7	15.6	18.1	17.5	13.5	11.5	8.1	5.5	(19)	
		MCWB	1.5	1.8	3.3	3.9	6.4	8.3	10.6	10.3	7.1	5.8	3.5	1.4	(20)	
	2%	DB	2.9	4.8	3.8	6.1	9.7	13.0	15.8	14.3	11.0	8.1	5.2	3.6	(21)	
		MCWB	0.0	0.2	1.1	2.7	5.3	7.9	9.9	9.1	6.0	3.6	2.2	1.4	(22)	
	5%	DB	1.3	2.5	2.0	4.1	7.7	11.3	14.2	12.6	9.4	6.1	3.2	1.9	(23)	
		MCWB	-1.0	-0.7	0.0	1.3	4.3	7.2	9.3	8.7	5.5	2.9	1.4	0.5	(24)	
	10%	DB	-0.3	0.0	0.4	2.4	6.0	9.7	12.8	11.3	8.1	4.7	1.8	0.1	(25)	
		MCWB	-1.9	-1.9	-1.5	0.2	3.5	6.7	8.9	8.3	5.0	2.1	0.4	-1.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.5	4.0	5.1	7.1	10.2	12.0	11.3	9.1	6.7	5.3	2.9	(27)	
		MCDB	4.0	5.6	6.8	6.8	11.0	12.3	15.6	14.2	12.2	11.0	6.7	3.5	(28)	
	2%	WB	0.7	0.9	1.6	3.1	5.9	8.4	10.7	10.1	7.2	4.6	3.1	1.9	(29)	
		MCDB	2.1	4.0	3.6	5.3	8.5	11.4	14.1	12.8	8.9	6.2	4.3	3.1	(30)	
	5%	WB	-0.8	-0.5	0.2	1.6	4.7	7.6	9.9	9.4	6.4	3.4	1.6	0.5	(31)	
		MCDB	1.2	2.0	1.8	3.6	7.2	10.5	13.1	11.8	8.1	5.2	2.6	1.7	(32)	
	10%	WB	-2.2	-2.0	-1.4	0.5	3.6	6.8	9.2	8.8	5.6	2.5	0.4	-1.0	(33)	
		MCDB	-0.1	0.6	0.2	2.1	5.6	9.4	12.0	10.8	7.3	4.2	1.6	0.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	4.6	4.8	4.5	4.5	4.9	5.2	4.4	3.7	3.6	3.4	3.3	(35)	
		MCDBR	5.3	6.9	6.0	6.2	7.4	8.2	8.1	6.9	5.6	5.2	5.1	5.0	(36)	
	5% WB	MCWBR	4.1	5.1	4.6	4.0	4.1	3.7	3.5	3.3	3.1	3.4	4.4	3.8	(37)	
		MCDWR	5.2	6.7	6.1	5.6	6.9	7.0	7.2	5.8	4.5	4.4	5.0	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.238	0.267	0.320	0.331	0.343	0.341	0.340	0.303	0.259	0.356	0.350	(40)	
		taud	N/A	2.272	2.364	2.282	2.352	2.425	2.475	2.484	2.512	2.329	2.828	2.914	(41)	
	Ebn at Noon		N/A	683	828	848	874	870	862	825	785	635	321	177	(42)	
		Edh at Noon	N/A	48	72	100	103	99	92	83	64	45	20	12	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.09	0.57	1.90	3.73	5.18	5.67	5.19	3.78	2.24	0.95	0.18	0.02	(44)	
	RadStd		0.01	0.06	0.18	0.29	0.35	0.49	0.50	0.26	0.25	0.08	0.03	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.73	N/A	N/A	+1.41	N/A	N/A	N/A	N/A	+11	N/A

Nomenclature: See separate page